

Pramaan (■■■■■■■■) — Project Documentation

Sovereign Government Document Intelligence & Agentic RAG Platform for India

1. Executive Summary

Pramaan is an AI-first document intelligence platform tailored for Indian government gazettes, ministry circulars, notifications, and statutory orders. It solves document fragmentation and bureaucratic complexity by combining automated sovereign web crawlers, hybrid PyMuPDF/GPT-4o Vision OCR layout extraction, high-dimensional vector embeddings, and a 6-stage Agentic RAG orchestrator.

2. System Architecture & 6-Stage Agentic RAG Pipeline

- **Stage 1: Query Understanding & Normalization** — Normalizes statutory identifiers and ministry taxonomy.
- **Stage 2: Orchestrator Agent** — Dynamic query routing and multi-hop planning.
- **Stage 3: Hybrid Retrieval** — Dense cosine similarity over pgvector (1536-dim) + sparse keyword filtering.
- **Stage 4: Multi-Document Reasoning** — Chronological diffing across fiscal years.
- **Stage 5: Evidence Validation** — Strict guardrails ensuring answers cite specific gazette clauses.
- **Stage 6: Response Generation** — Citations mapped directly to page, section, and clause.

3. Ingestion & Storage Tiering

- **Storage Layer:** PostgreSQL 15 + pgvector with HNSW index for sub-15ms cosine retrieval.
- **Extraction Engine:** PyMuPDF for native digital PDFs + OpenAI GPT-4o Vision for scanned stamped physical gazettes.
- **Active Crawler:** Automated Celery queue polling egazette.gov.in and pib.gov.in with live UI sync triggers.

4. Verified Deliverables & Interface

- **Minimalist Workspace (/app):** ChatGPT-inspired layout with collapsible navigation and slide-in Evidence Drawer.
- **Compare Tool:** Statutory diff engine for legislative amendments.
- **Documents Library:** Search, filter, and on-demand portal crawler sync.